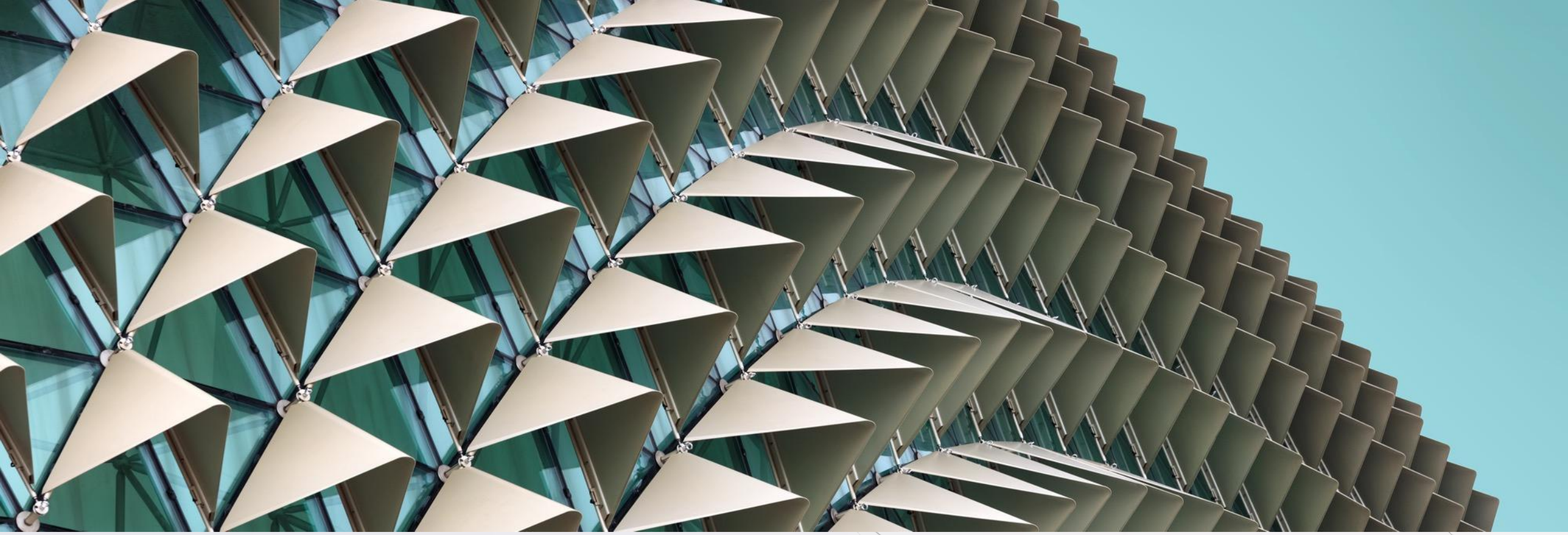

DP WORLD

Snack Overflow

TEAM - DPWH007

PROJECT TITLE AND TEAM INTRODUCTION

Project Title:
Port Brain: An AI-Driven Maritime Decision Intelligence and Port Docking Optimization System

Team Introduction:

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Team Github Repository:

https://github.com/Shivansh-00/DPWH007

PROBLEM STATEMENT

- **Port Congestion & Delays:** Ships arriving at ports often face long waiting times due to inefficient scheduling and berth allocation. This leads to congestion in anchorage zones and delays in cargo handling.

- **Fuel & Cost Wastage:** Idle ships consume fuel while waiting, increasing operational costs and reducing profitability. This also impacts shipping schedules and downstream logistics.

- **Lack of Intelligent Decision Systems:** Most ports rely on static or manual decision-making instead of dynamic, data-driven systems. This results in suboptimal berth utilization and poor traffic control.

- **Handling Diverse Ship Types:** Different ships (container, tanker, bulk, food) have varying priorities and requirements. Current systems fail to balance urgency, safety, and efficiency effectively.

Target Customers include Port authorities, logistics companies, and maritime operators. This will also be useful for smart port development initiatives.

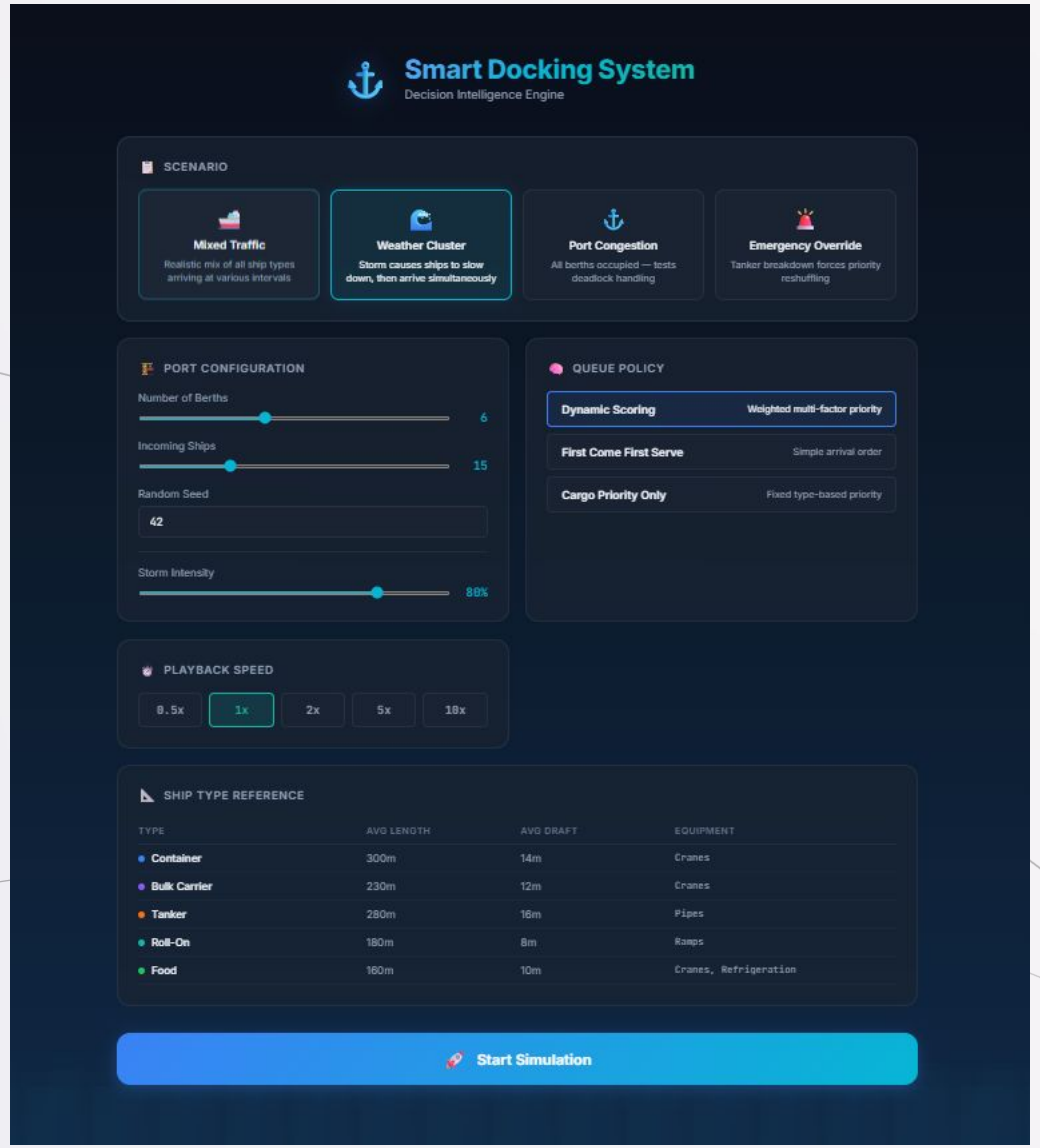
PROPOSED SOLUTION/SOLUTION OVERVIEW

1. **AI-Powered Port Decision System:** Port Brain is a smart system that dynamically assigns berths and controls ship entry using real-time data. It acts as a central decision engine for port operations.

2. **Zone-Based Traffic Control Model:** The system divides the port into zones: Open Sea, Anchorage, Entry Channel, and Berths. This ensures realistic and controlled ship movement.

3. **Dynamic Ship Labeling & Tracking:** Ships are identified when they cross a defined boundary and continuously tracked. Their state transitions are managed throughout the docking process.

4. **Dynamic Priority Scoring System:** Ships are prioritized using multiple factors like cargo type, ETA urgency, waiting time, and risk conditions. This ensures fair and adaptive decision-making.

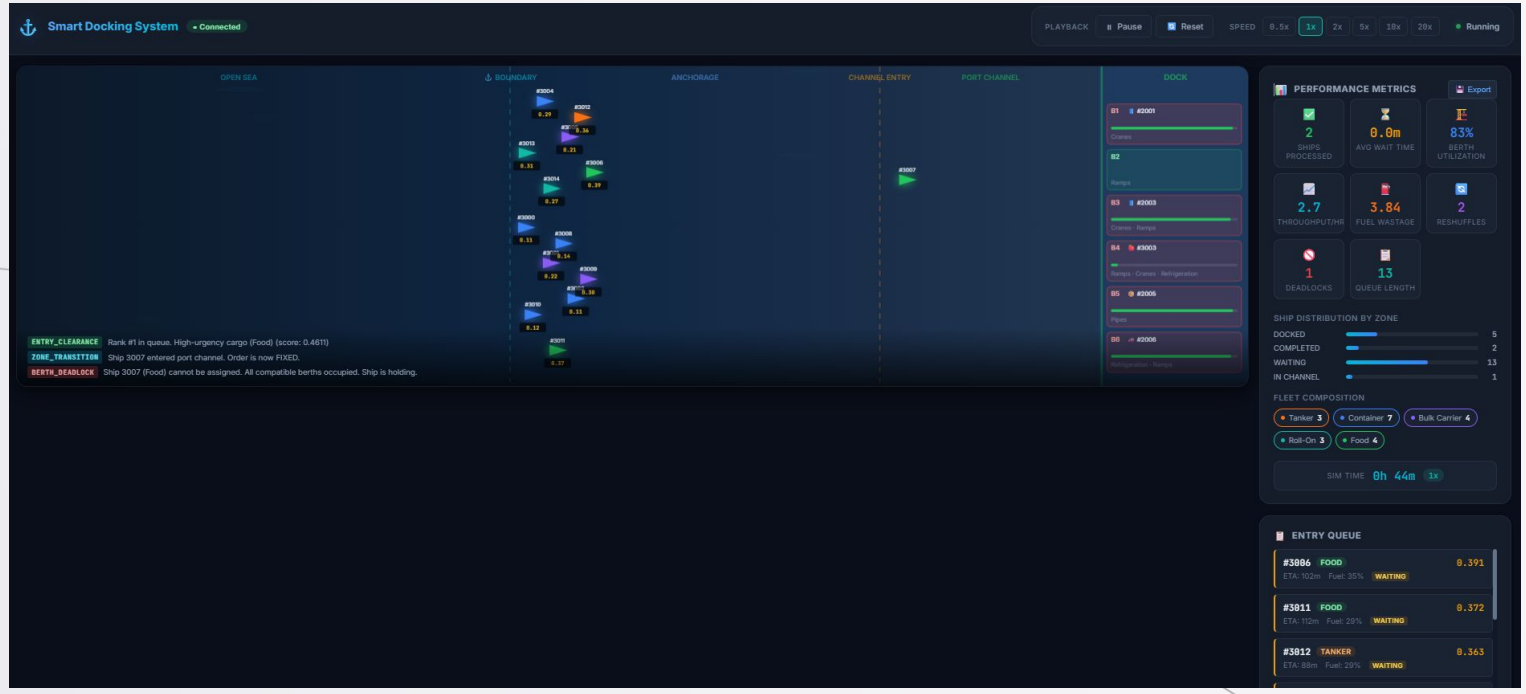


PROPOSED SOLUTION/SOLUTION OVERVIEW

5. **Controlled Entry Sequencing:** Instead of overtaking, ships are held or cleared at anchorage zones. Entry into the channel is strictly controlled for safety and realism.

6. **Optimized Berth Assignment:** Berths are assigned based on compatibility, availability, and estimated turnaround time. The goal is to minimize total docking and unloading time.

7. **ETA Prediction with Risk Factors:** Arrival time is predicted using speed, distance, and external risks like weather or congestion. This helps ports prepare for incoming traffic.



8. **Scenario-Based Simulation Engine:** The system simulates events like storms, congestion waves, and delays. This enables testing of real-world situations.

9. **What-If Analysis Mode:** Different strategies (FCFS, priority-based, scoring-based) can be compared. This helps evaluate and improve decision policies.

TECHNOLOGY STACK

1. **Frontend:**

   Next.js with Canvas-based simulation to visualize ship movement and docking.

2. **Backend:**

   FastAPI handles simulation logic, APIs, and Websocket communication. It processes real-time decisions simulations and updates.

3. **Database:**

   MongoDB stores AIS data, simulated states, and events. Its flexible structure supports dynamic simulation data.

4. **AI/ML Components:**

   ETA prediction and congestion estimation models enhance decision-making.

5. **Real-TIme Processing:**

   Websockets stream live ship states and decisions to the UI. Ensures smooth and continuous simulated updates.

6. **Data Sources:**

   Uses Marine Cadastre AIS Dataset for realistic ship movement data. Anomaly injection adds real-world complexity.

7. **Deployment:**

   Docker

8. **Security:**

   JWT Authentication and Role-based Access

BASIC PROTOTYPE

Our solution aligns strongly with DP World's core businesses:

1. Port Terminal Operations: Optimizes berth allocation and ship sequencing, reducing congestion and turnaround time.
2. Cargo Logistics: Ensures faster and more predictable cargo flow, improving supply chain efficiency.
3. Maritime Services: Enhances vessel traffic management through intelligent, data-driven decision-making.
4. Free Trade Zones: Supports high-throughput operations by minimizing delays and improving port productivity.

Overall, it helps DP World improve efficiency, reduce costs, and move towards smarter, AI-driven port operations.

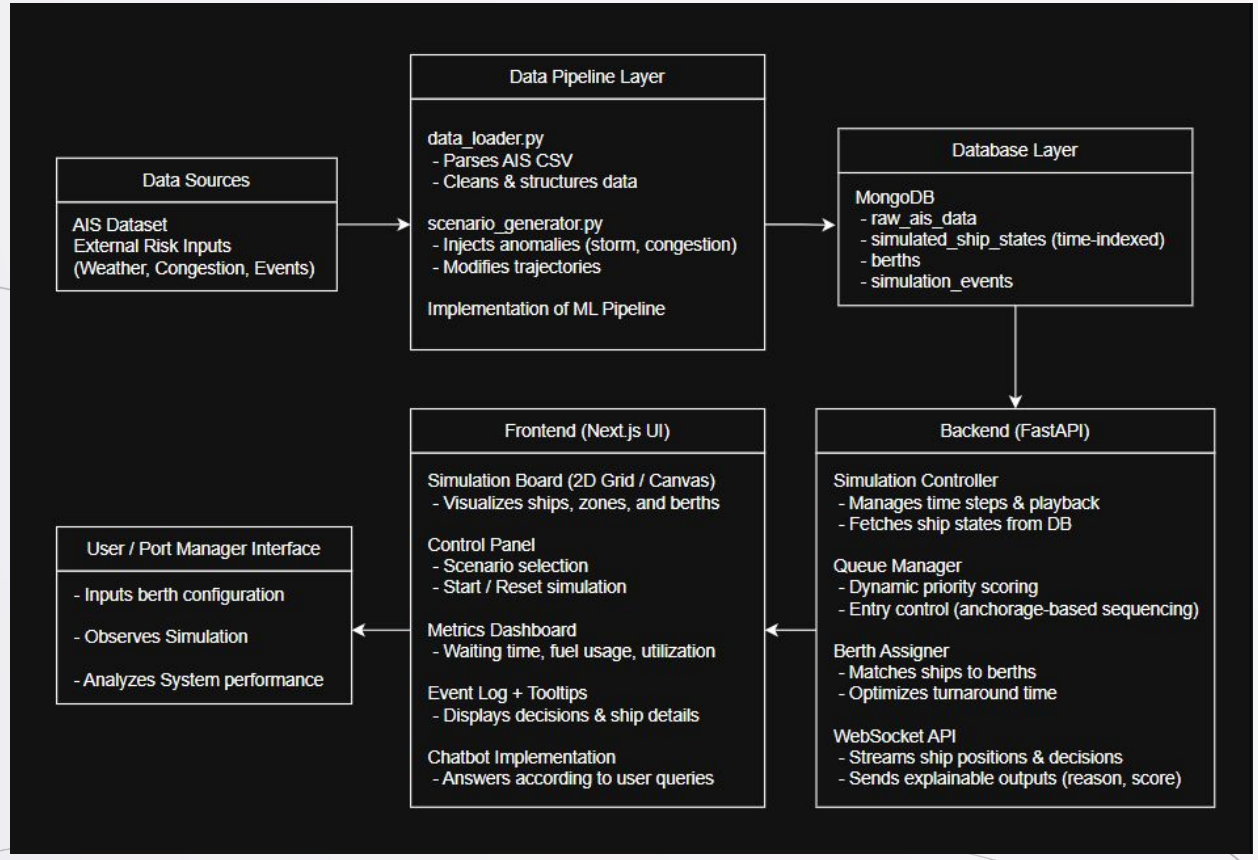


Fig. Architecture Diagram for Working of Port Brain

MARKET ANALYSIS

1. **Target Audience:**
   Port authorities, logistics companies, and maritime operators managing global cargo flow.
   Increasing demand from smart port and digital transformation initiatives.

2. **Growing Maritime Trade:**
   Over 80–90% of global trade is carried by sea, and cargo volume is expected to grow steadily in the coming years. Ports must scale operations efficiently to handle rising demand.

3. **Rising Congestion & Costs:**
   Port congestion can increase operational costs by 20–30% due to delays, fuel wastage, and inefficiencies. This creates a strong need for optimization systems.

4. **Shift Towards AI & Smart Ports:**
   The global smart ports market is projected to grow significantly (estimated 20%+ CAGR).
   AI-based decision systems are becoming key investments.

BUSINESS CASE

Cost and Operational Improvements:

Reduced Waiting Time: Efficient scheduling reduces idle time for ships. This improves overall port throughput.
Fuel Cost Savings: Less waiting means lower fuel consumption. This directly reduces operational expenses.
Better Resource Utilization: Optimized berth allocation improves infrastructure usage. Ports can handle more ships without expansion.

Financial & Sustainability Impact:

Increased Revenue Opportunities: Faster turnaround allows ports to handle more ships. This increases revenue generation.
Reduced Environmental Impact: Lower fuel usage reduces emissions. This supports sustainable maritime operations.

Improved Supply Chain Efficiency: Timely docking ensures faster cargo movement. This benefits global trade networks.

INNOVATIVE ASPECTS

Unique Selling Point of Port Brain:

- **Early Labelling & Proactive Decision System:** Ships are identified as soon as they enter a defined boundary, allowing decisions to be made before congestion occurs, not after.

- **Multi-Type Priority Intelligence:** The system differentiates between food, tanker, Ro-Ro, container, and bulk ships, assigning priority based on real-world urgency and risk.

- **Realistic Anchorage-Based Control Model:** EInstead of unrealistic reshuffling, ships are managed through controlled entry from anchorage zones, aligning with real maritime operations.

- **Explainable Decision Engine:** Every decision is explained, such as why a ship is delayed or prioritized, making the system transparent, trustworthy and easy to analyze.

- **Chatbot Integration:** Answers queries asked by users / Port Manager.

THANK YOU